



RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY

PRESENTATION ON

LEVERAGING TRANSFORMER BASED MODEL FOR ENHANCED HYPERSPECTRAL IMAGE CLASSIFICATION

Presented by:

Rafi Rafsan

Roll: 1903139

Supervised by:

Dr. Boshir Ahmed

Professor

Dept. of Computer Science &
Engineering, RUET

CONTENTS

- ★ Introduction
- ★ Literature Review
- ★ Challenges
- ★ Objective
- ★ Methodology
- ★ Data Description
- ★ Research Implementation
- ★ Conclusions
- ★ Future Work
- ★ References

INTRODUCTION

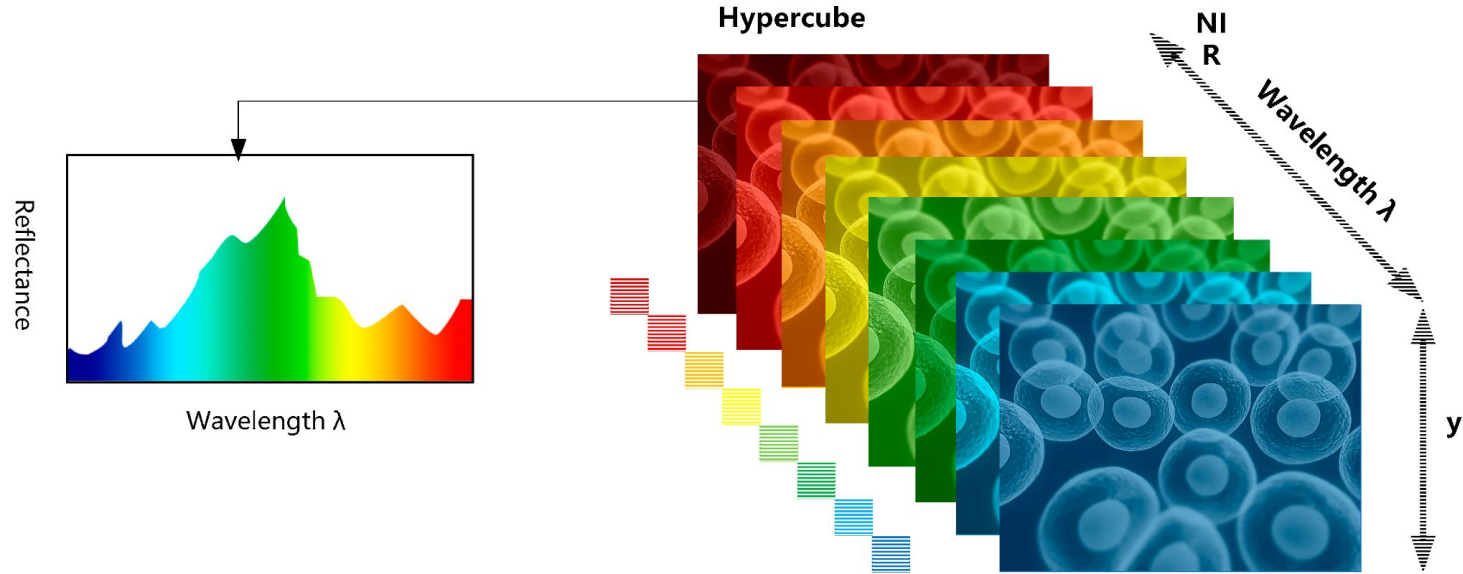


Figure 1: Spectral signature of HSI

Source: Cui et al., Sensors, 2022

BENEFITS OF HYPERSPPECTRAL IMAGING

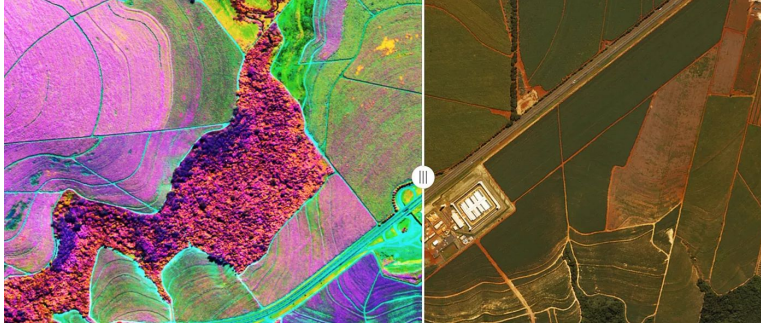


Figure 1: Precision agriculture using HSI

- Detect stressed crops
- Identify nutrient deficiencies
- Optimize resource allocation

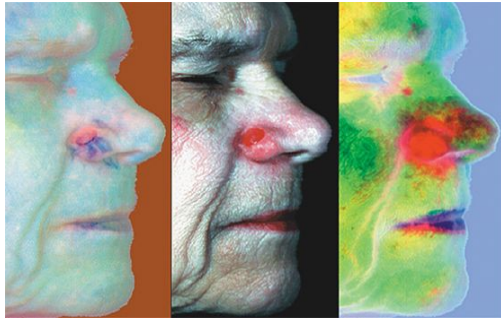


Figure 2: Cancerous tissues detection

- Identifying **cancerous tissues** (e.g., skin, colon, brain tumors)
- Detecting **diabetic ulcers** or **retinal diseases**

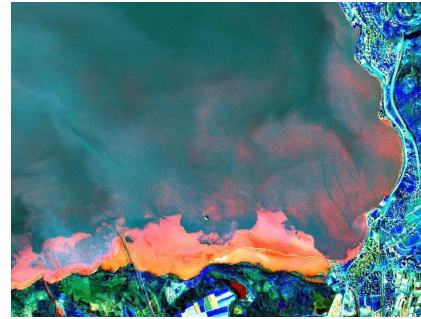


Figure 3: Natural resource management

- Mitigate environmental hazards more efficiently
- Protect ecosystems
- Conserve biodiversity

LITERATURE REVIEW

No.	Study	Key Contribution	Limitations
1.	CTF-SSCL (CNN-Transformer with Semi-Supervised Contrastive Learning)	• Combines CNN for local Transformer for global feature extraction • Uses Semi-Supervised Contrastive Learning (SSCL) • Introduces Spectral Feature Shift Strategy (SFSS)	• CNNs have limited receptive field (weak global context) • Transformers often neglect local spatial-spectral details
2	DCFSL (Deep Cross-Domain Few-Shot Learning)	• First to integrate Few-Shot Learning with Domain Adaptation • Uses Conditional Adversarial Domain Adaptation (CDAN) • Learns transferable features across domains	• Assumes class consistency between source and target • Requires significant labeled source data

1. Wang, X., Guo, J., Liu, X., & Zhong, P. (2023). CTF-SSCL: CNN-Transformer for Few-Shot Hyperspectral Image Classification with Semi-Supervised Contrastive Learning. *IEEE Transactions on Geoscience and Remote Sensing*, 61, 1–15. <https://doi.org/10.1109/TGRS.2023.3245678>

LITERATURE REVIEW

No.	Study	Key Contribution	Limitations
3.	Residual Dense Connection Network with Transfer Learning)	• Applies Meta-Transfer Learning for low-data adaptation • Employs 3D residual + dense blocks for strong spectral-spatial feature extraction	• Dependent on pretraining • Shallow classifiers may miss deep semantic features

2. Zhang, X., Li, Y., & He, Y. (2022). Deep Cross-Domain Few-Shot Learning for Hyperspectral Image Classification. ISPRS Journal of Photogrammetry and Remote Sensing, 188, 32–44. <https://doi.org/10.1016/j.isprsjprs.2022.04.005>
3. Zhou, H., Wang, X., Xia, K., Ma, Y., & Yuan, G. (2024). Transfer Learning-Based Hyperspectral Image Classification Using Residual Dense Connection Networks. Sensors, 24(9), 2664. <https://doi.org/10.3390/s24092664>

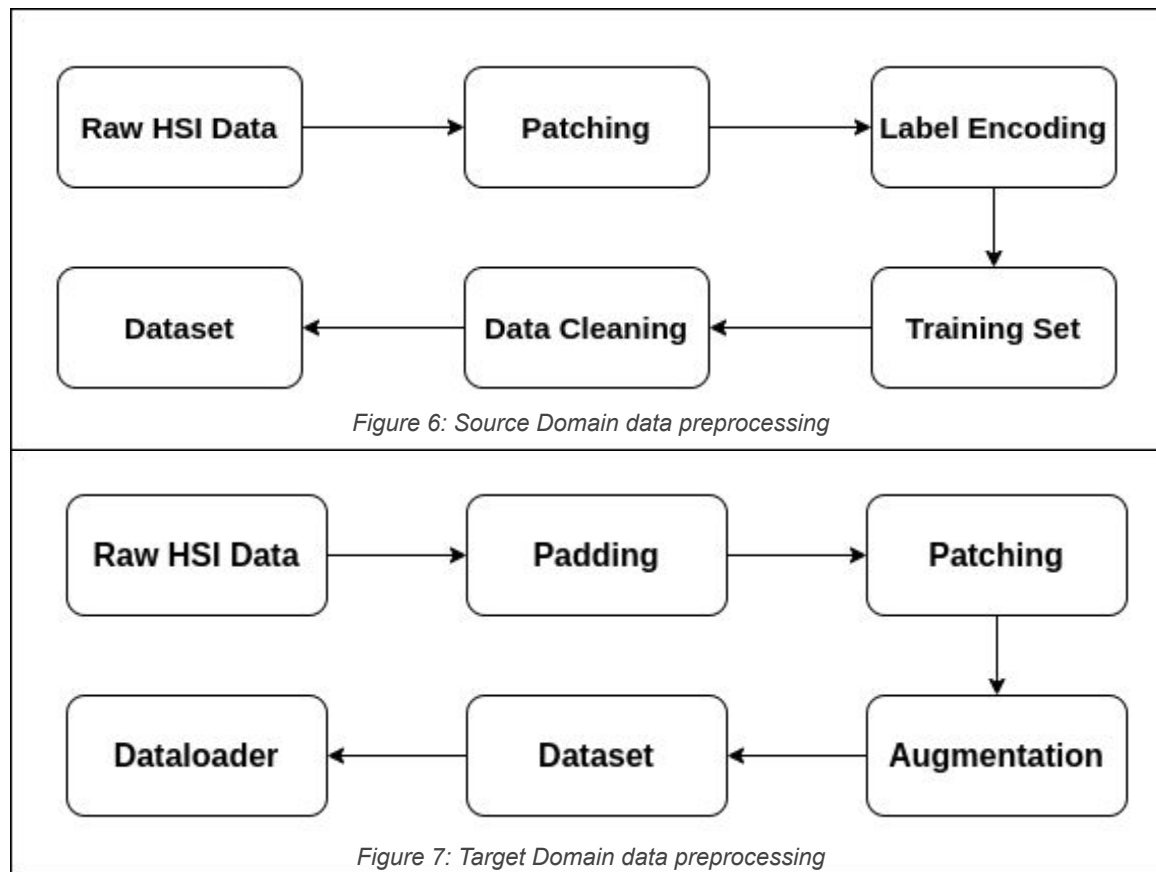
CHALLENGES

- Scarcity of labeled hyperspectral data
- Domain shift across regions/sensors
- Ineffective feature representation

OBJECTIVES

- Develop a transformer-based few-shot learning framework.
- Enhance cross domain generalization using domain adaptation.
- Validate the model's ability to generalize to unseen target domains with minimal labeled data

PREPROCESSING STEPS



METHODOLOGY(WORKFLOW)

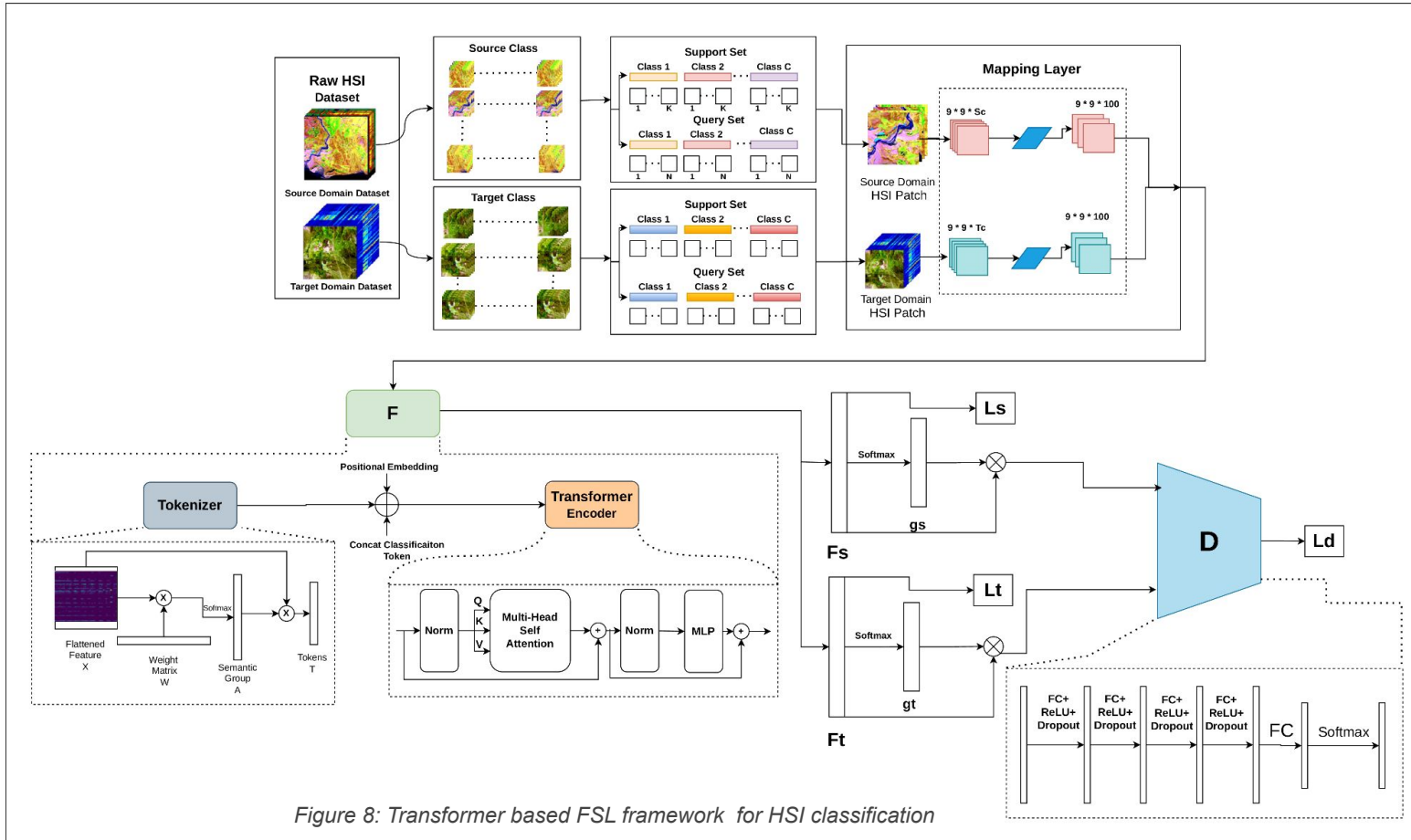


Figure 8: Transformer based FSL framework for HSI classification

METHODOLOGY (CONT'D)

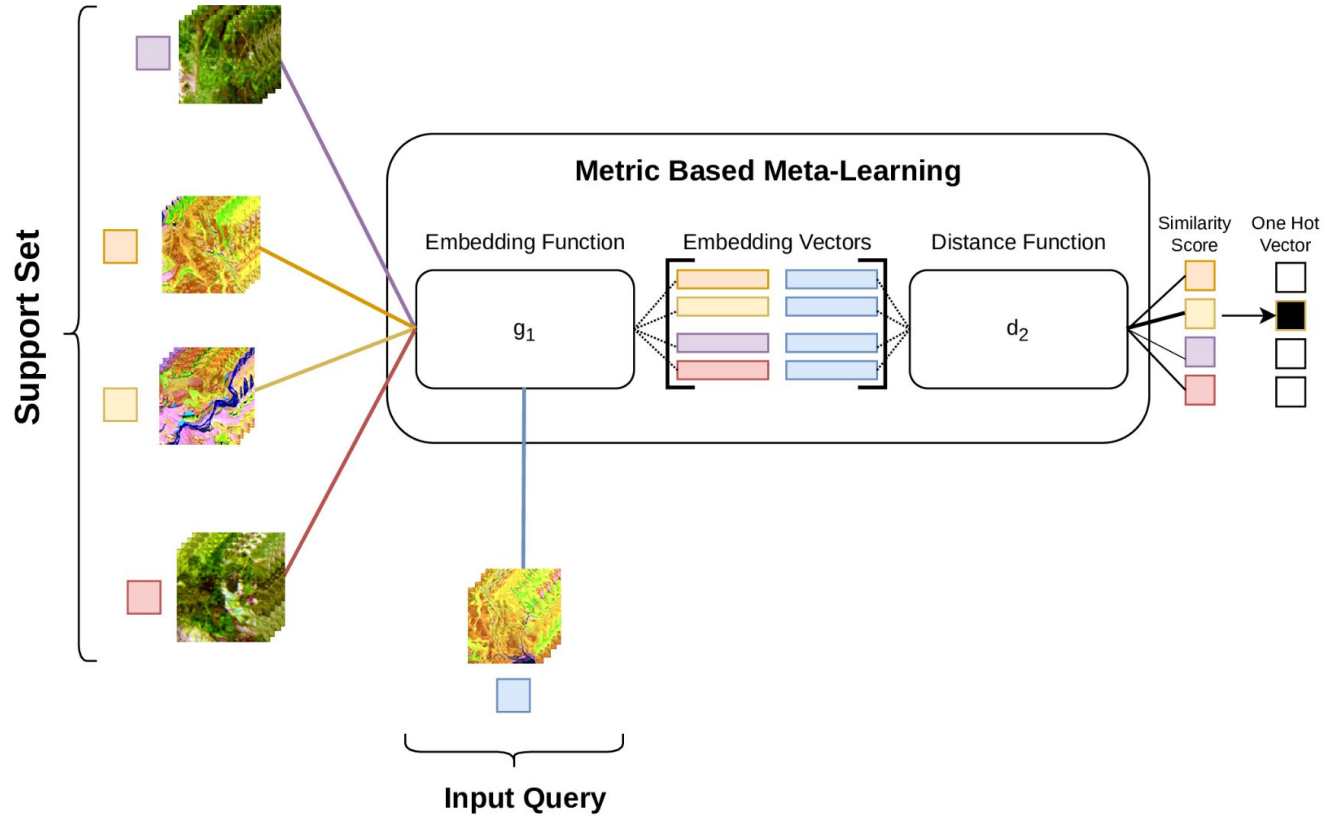


Figure 9: Meta learning (Adapted from [Parnami & Lee, 2022])

Leveraging Transformer-Based Models for Enhanced Hyperspectral Image Classification

METHODOLOGY (CONT'D)

CONDITIONAL DOMAIN ADAPTATION

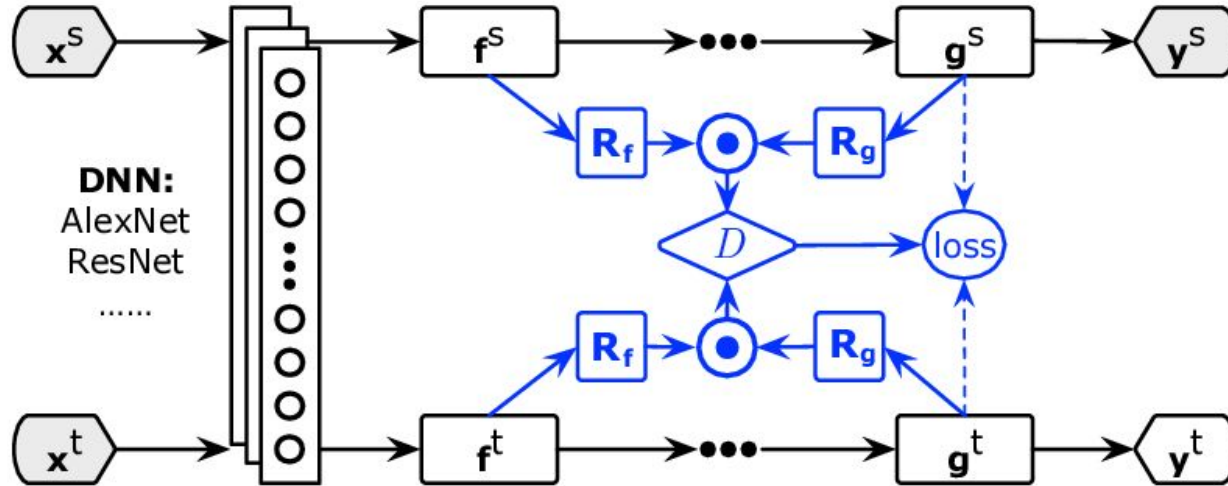


Figure 10: Randomized Multilinear Conditioning (Adapted from [Long et al., 2018])

METHODOLOGY (CONT'D)

MODEL EVALUATION ON TARGET DOMAIN

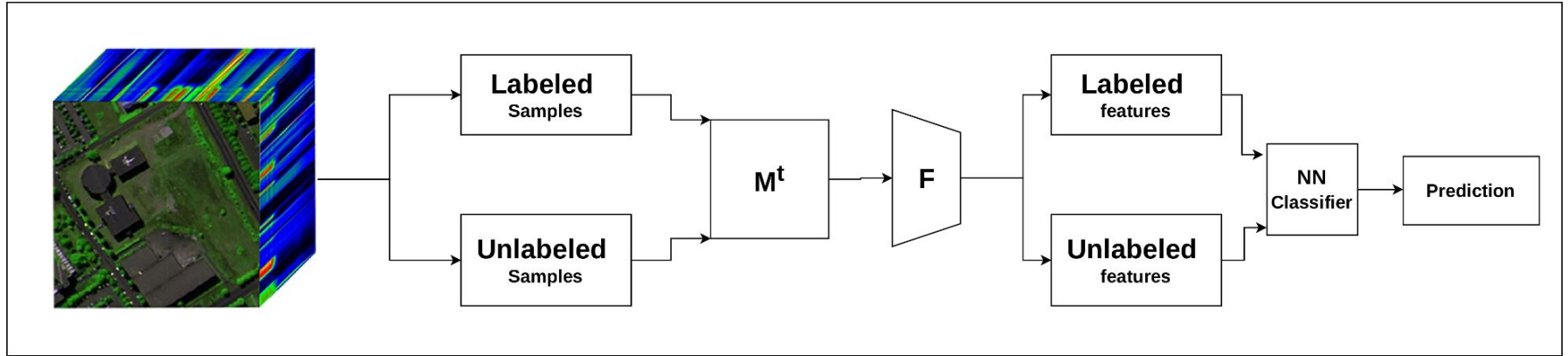


Figure 10: Testing phase

DATA DESCRIPTION

Dataset	Source	Spectral Band	Spatial Resolution	Classes
Chikusei(Source)	Airborne sensor	128 (363–1018 nm)	2.5 m	Urban, Vegetation
PaviaU(Target)	ROSIS-03 sensor	103 (430–860 nm)	1.3 m	Urban areas

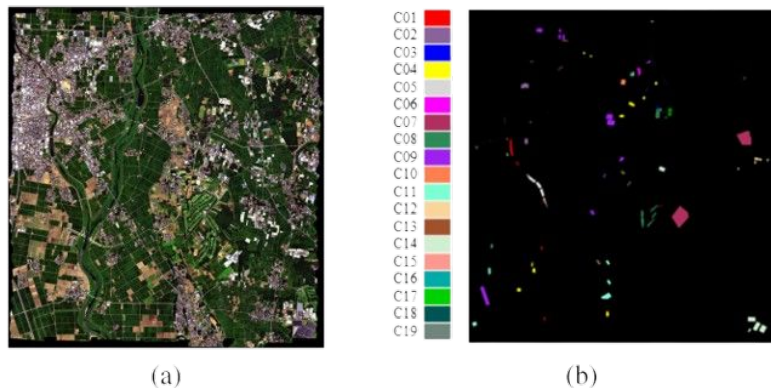


Figure 4: (a) Chikusei Dataset. (b) Ground-truth map

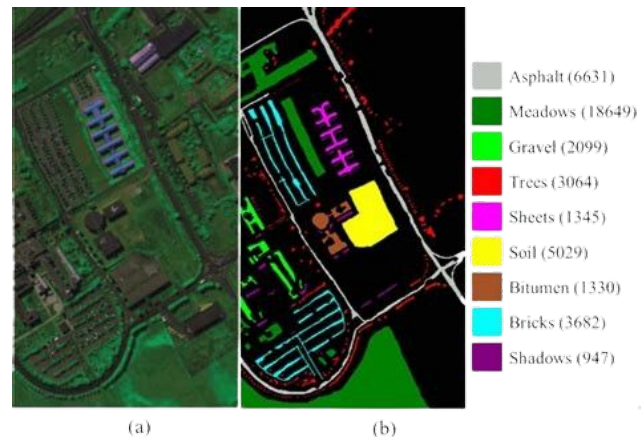


Figure 5: (a) PaviaU Dataset. (b) Ground-truth map

RESEARCH IMPLEMENTATION

- Training: 10,000 episode (9 way 5 shot)
- Evaluation: Accuracy, Kappa coefficient.
- Visuals:
 - t-SNE plots
 - Confusion matrix
 - Per-class accuracy

TRAIN LOSS

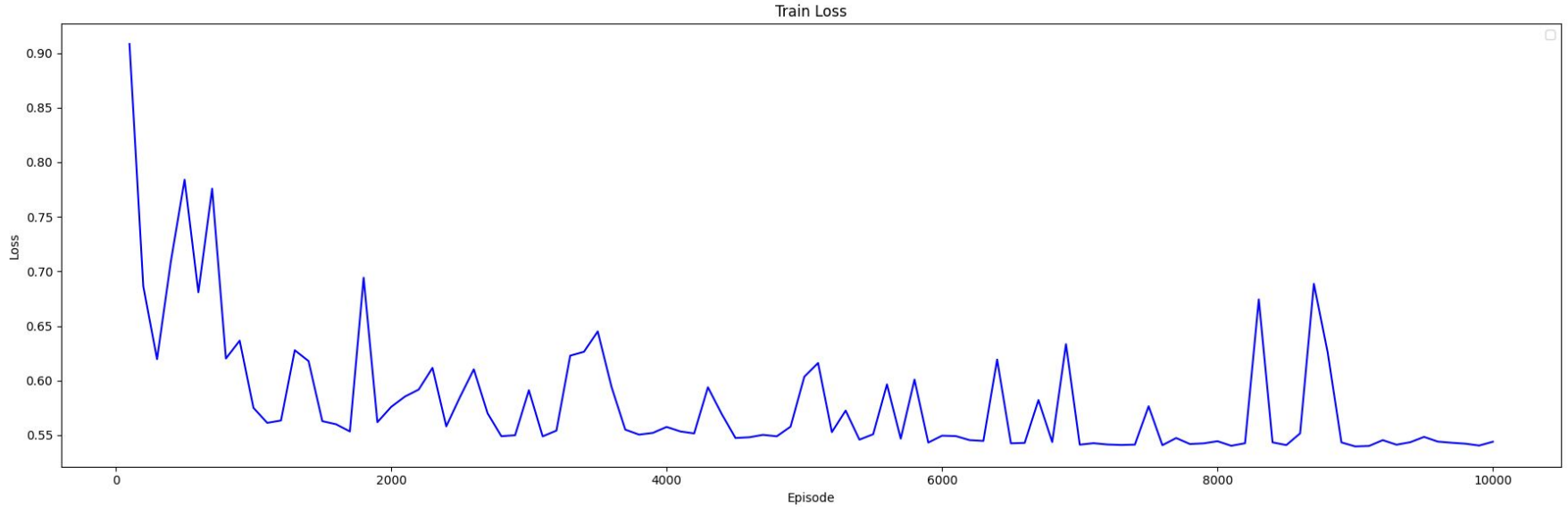


Figure 11: Train loss curve on PaviaU dataset

FEATURE VECTOR VISUALIZATION

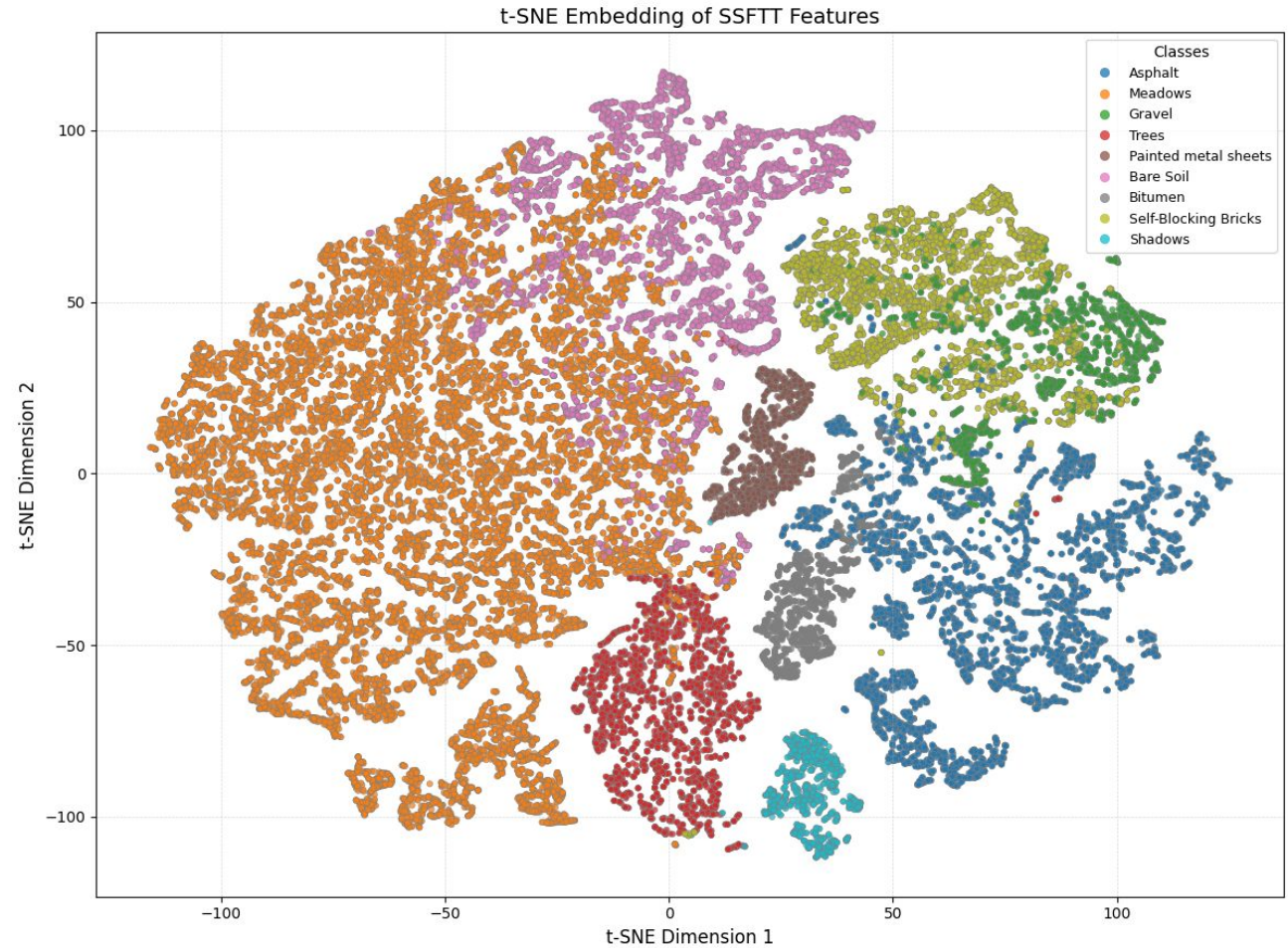


Figure 12: Visualization of embedded feature vectors by t-SNE maps on UP dataset

EVALUATION ACROSS FEW-SHOT SAMPLE

Performance Metrics Across Few-Shot Scenarios

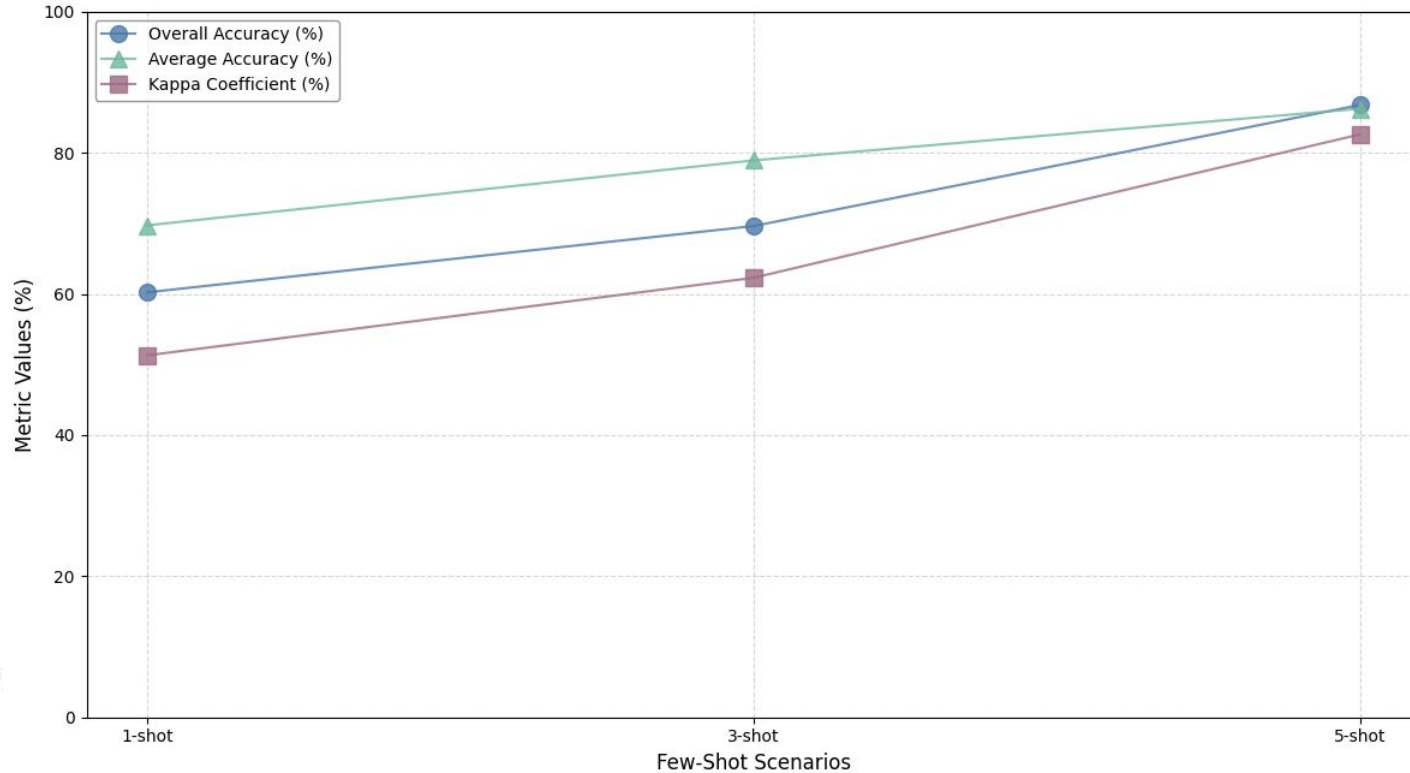


Figure 13: Different few-shot scenarios



CLASSIFICATION MAP

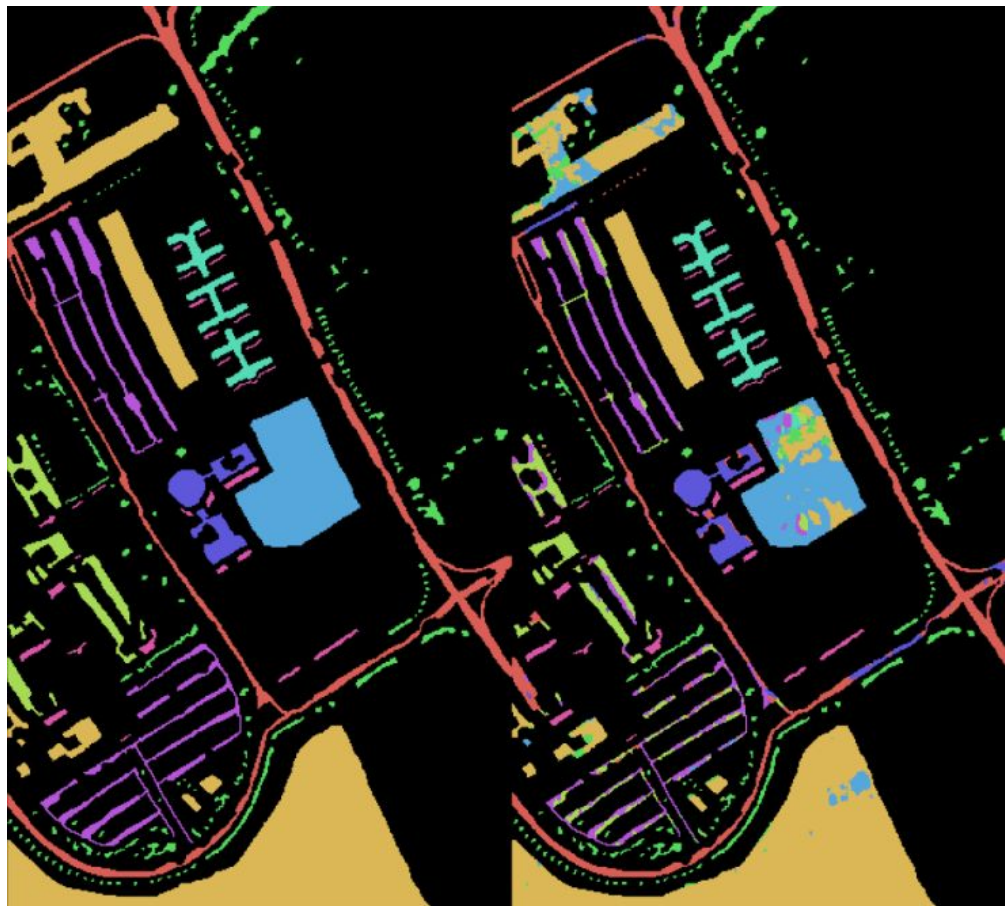


Figure 14: Classification map of PaviaU

CONFUSION MATRIX

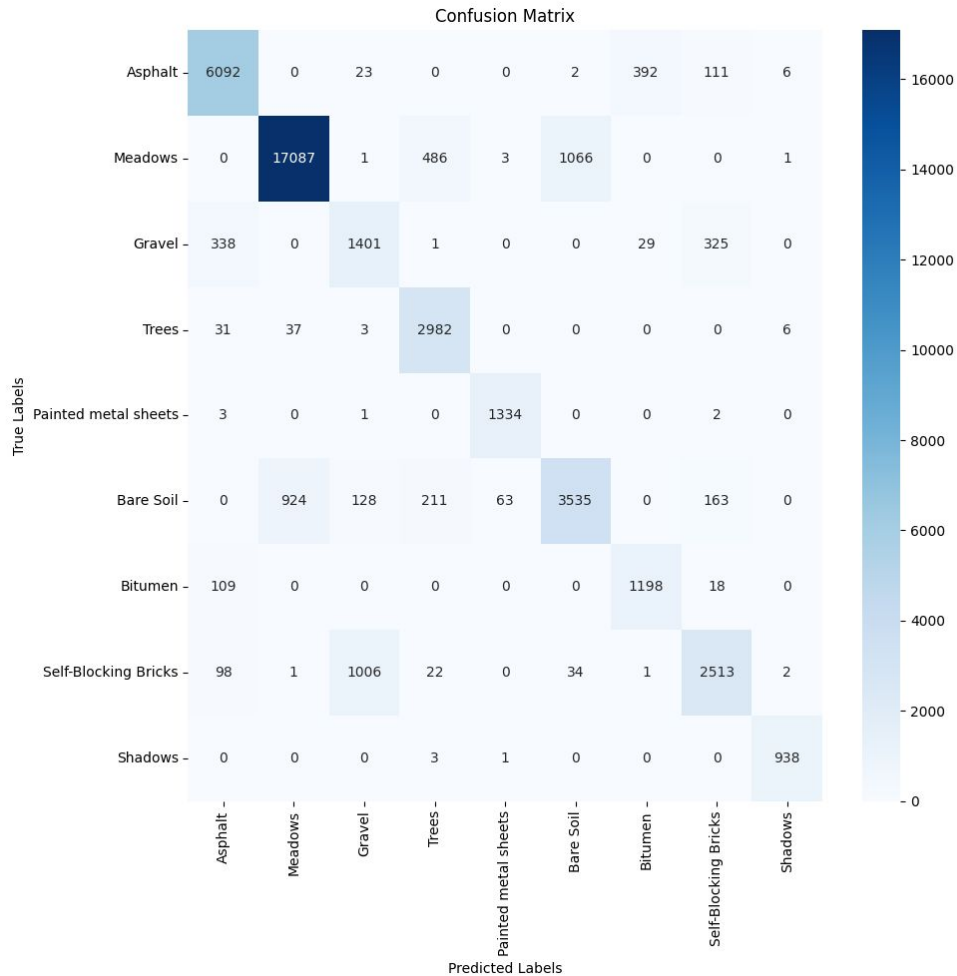


Figure 15: Confusion Matrix

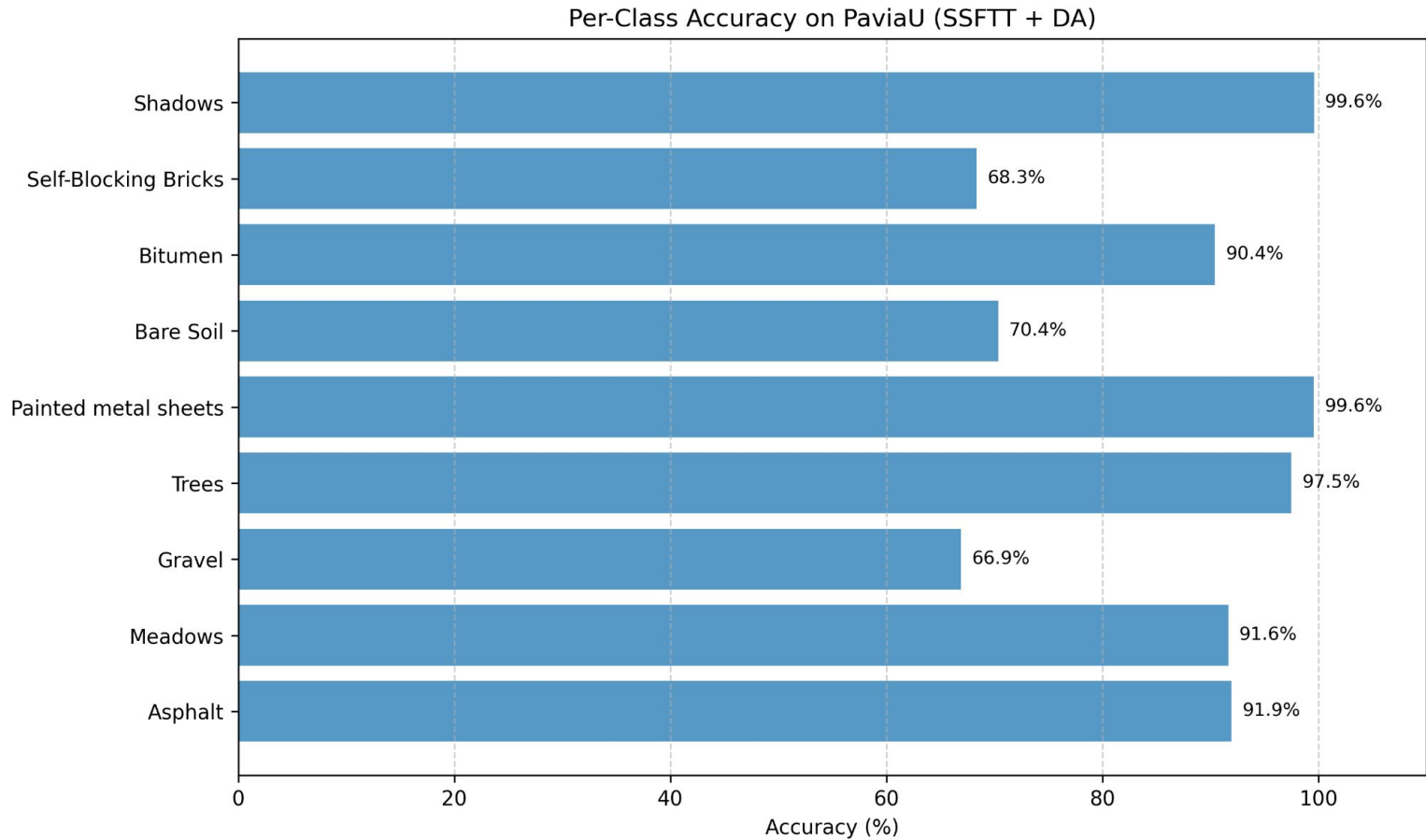


Figure 16: Per class accuracy

COMPARATIVE PERFORMANCE

Model	Overall Accuracy (%)	Average Accuracy (%)	Kappa (%)
MFSC	86.02	88.21	81.93
DCFSL	83.65	83.77	78.70
CTF-SSCL	82.90	85.67	77.99
Ours (w/o DA)	79.60	80.30	74.12
Ours(w/ DA)	86.78	86.25	82.61

CONCLUSION

- Outperformed existing few-shot models on overall accuracy and kappa coefficient.
- Better generalization with domain adaptation
- Effective under extreme label constraints.

FUTURE SCOPE

- Incorporate unsupervised or self-supervised pre training
- Enhance prototype learning for novel class generalization
- Investigate tokenization strategies for variable-length spectral inputs

REFERENCES

- Zhou, H., Wang, X., Xia, K., Ma, Y., & Yuan, G. (2024). Transfer Learning-Based Hyperspectral Image Classification Using Residual Dense Connection Networks. *Sensors*, 24(9), 2664. <https://doi.org/10.3390/s24092664>
- L. Sun, G. Zhao, Y. Zheng and Z. Wu, "Spectral–Spatial Feature Tokenization Transformer for Hyperspectral Image Classification," in *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1-14, 2022, Art no. 5522214, doi: 10.1109/TGRS.2022.3144158.
- Z. Li, M. Liu, Y. Chen, Y. Xu, W. Li and Q. Du, "Deep Cross-Domain Few-Shot Learning for Hyperspectral Image Classification," in *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1-18, 2022, Art no. 5501618, doi: 10.1109/TGRS.2021.3057066.
- B. Liu, X. Yu, A. Yu, P. Zhang, G. Wan and R. Wang, "Deep Few-Shot Learning for Hyperspectral Image Classification," in *IEEE Transactions on Geoscience and Remote Sensing*, vol. 57, no. 4, pp. 2290-2304, April 2019, doi: 10.1109/TGRS.2018.2872830.
- M. Long, Z. Cao, J. Wang, and M. I. Jordan, "Conditional adversarial domain adaptation," *Advances in Neural Information Processing Systems*, vol. 31, pp. 1640–1650, 2018.

Q&A

- Thank you for your attention. Any questions?